

Laila Mohamed Fikry

Badr City, Cairo, Egypt

laila.mohamed.fikry@gmail.com | +20 121 021 2792 | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Professional Summary

Full Stack Developer and Software Engineer currently in Year 3 of a Computer Science degree, with production experience most candidates do not reach until years into their careers. Built and deployed enterprise IoT systems serving the Ministry of Interior (MOI) and the Egyptian Natural Gas Holding Company (GASCO), trained deep learning models for real-world emergency response, and shipped full-stack web applications for paying clients – all while studying. Proficient in C#, ASP.NET, Python, React.js, and REST API design across the full software development lifecycle (SDLC). Actively seeking a software engineering internship where I can contribute immediately and keep growing fast.

Technical Skills

- **Programming Languages:** C#, C, C++, Python, Java, JavaScript, SQL, Bash
- **Backend Development:** ASP.NET, Node.js, Express.js, RESTful API Design, Custom Windows Services
- **Frontend Development:** React.js, Next.js, Tailwind CSS, HTML5, CSS3, JavaScript
- **Artificial Intelligence and Machine Learning:** TensorFlow, Keras, Convolutional Neural Networks (CNN), Transfer Learning, MobileNetV2, Grad-CAM, Computer Vision, Deep Learning
- **IoT and Protocols:** SNMP, Modbus, HTTP, Hardware Integration, Industrial Device Communication
- **Databases:** SQL Server, PostgreSQL, MySQL, MongoDB
- **Tools and Practices:** Git, Version Control, MSBuild, Linux (Ubuntu), Bash Scripting, Agile, Scrum, Software Development Lifecycle (SDLC)

Work Experience

LM Tech Solutions

Lead Software Engineer

Cairo, Egypt

Nov 2025 – Present

- **Architected RMS 3.0 Enterprise IoT Platform:** Led end-to-end software development of the Remote Monitoring System (RMS 3.0), an industrial Internet of Things (IoT) platform for critical power infrastructure including rectifiers, solar systems, inverters, and rectiverters, with a scalable architecture designed for future universal device integration
- **Engineered Unified Polling Service:** Built a fault-tolerant backend service in C# and ASP.NET using MSBuild to handle concurrent device communication and real-time data ingestion across Modbus, HTTP, and SNMP protocols, following object-oriented programming best practices
- **Led Production Enterprise Deployments:** Overseeing testing and production rollout of RMS 3.0 for major national clients including the Ministry of Interior (MOI) and the Egyptian Natural Gas Holding Company (GASCO), enhancing SQL Server database architecture to meet strict enterprise specifications
- **Developed Real-Time Operational Dashboards:** Built a scalable full-stack frontend using React.js and Tailwind CSS to visualize live power metrics, providing Network Operations Center (NOC) teams with visual alerts, condition reports, and performance tracking

Media Gate Company

Freelance Full Stack Developer

Cairo, Egypt

August 2025 – Oct 2025

- **Engineered Payment and Admin Workflows:** Integrated the Stripe API to process subscription payments for [bagijob.com](#), and built a comprehensive admin dashboard with automated revenue calculations and real-time approval notifications
- **Optimized Media Delivery Performance:** Accelerated video upload speeds and delivery by integrating Bunny Content Delivery Network (CDN), and improved frontend performance through custom scroll-aware video autoplay and pause functionality
- **Resolved Complex State Management Issues:** Fixed persistent dark mode preferences across sessions, standardized responsive ad card layouts, and built dynamic redirect flows for mobile and desktop payment outcomes

ALX Africa

Software Engineering Intensive Training Program

Remote
October 2023 – July 2025

- Completed a 15-month intensive full-stack development and systems engineering program covering the complete software development lifecycle
- Built and deployed 30+ production-style projects using Python, C, JavaScript, and SQL with version control via Git
- Designed RESTful APIs and full-stack web applications using React.js, Node.js, Express.js, and MongoDB
- Worked extensively in Linux environments using Bash scripting and Agile and Scrum workflows

Projects

Inqaz-app – Egypt Emergency AI Response System

GitHub

Full-Stack AI Platform – Python, Computer Vision, REST API, Web Frontend

Jan 2026

- Architected an end-to-end emergency response platform using computer vision to detect accidents and disasters from live mobile camera footage, classify scene severity, and automatically dispatch emergency services in real time
- Designed and built a full-stack web interface for live camera ingestion, AI-powered incident triage, and GPS-coordinated automated dispatch alerts to Ministry of Interior emergency line (122) and Ambulance services (123)
- Led all system architecture decisions across the computer vision pipeline, backend REST API, and frontend dispatch dashboard – full scope and technical detail available on GitHub

Crash Detection and Classification Model

GitHub | Live Demo

Deep Learning, CNN, Transfer Learning, Grad-CAM, Python, Streamlit

Dec 2025

- Trained two deep learning models on 3,000 real-world traffic images: a custom Convolutional Neural Network (CNN) built from scratch, and a MobileNetV2 transfer learning model achieving 68% F1-score on entirely unseen test data
- Applied Grad-CAM explainability to generate thermal heatmaps identifying exact regions of vehicle structural damage detected by the model, making AI decisions interpretable for emergency operators
- Deployed a production-ready web application on Streamlit Community Cloud with live camera ingestion, real-time classification, metric tracking, and simulated emergency dispatch – accessible at the live demo link above

Secure Real-Time Chat System

Dec 2025

Python, TCP Sockets, AES-256 Encryption, Multi-threading

- Developed a real-time encrypted messaging system using TCP sockets with multi-user support and a multi-threaded server architecture
- Implemented AES-256 encryption for end-to-end message security and SHA-256 hashing for user authentication with persistent data storage

Tech-Road – Workforce Readiness Platform (Digitopia Competition)

August 2025

Full-Stack Web Development, System Architecture, REST API Design

- Built a full-stack web platform bridging academic learning with real-world workforce requirements, developed under competition conditions
- Served as Team Leader and Backend Developer, responsible for overall system architecture, REST API design, and backend implementation

Education

Bachelor of Science in Computer Science, Expected Graduation 2027

El Sewedy University of Technology – Polytechnic of Egypt

2025 – Present

Egyptian Russian University

2023 – 2025 (Transferred)

- Relevant Coursework: Data Structures and Algorithms, Operating Systems, Software Engineering, Artificial Intelligence, Object-Oriented Programming

Volunteering

Computality Community – HR Team

Cairo, Egypt

2023 – 2024

- Conducted interviews, managed records for 60+ members, and organized university technology events

Languages

- Arabic (Native) | English (Fluent) | Turkish (Proficient)